



Théophile GOUSSELOT

Graduate Engineer and PhD Student in
microelectronics, computer science and security

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Aix-en-provence
Driving licence
26 years



I am looking for a challenging position that will stimulate me to develop, attack or protect chips for hours.
I am used to collaborating, presenting and teaching with enthusiasm!

Education

• PhD Student at Mines de St-Etienne since jan. 2022

Protection of RISC-V processors against fault injections

- **3rd best Hardware Demo (among 24)** at IEEE International Symposium on Hardware Oriented Security and Trust (HOST), 2023, USA, California, San Jose
- **[Paper]** Code Encryption for Confidentiality and Execution Integrity down to Control Signals, HOST, 2025
- **[Paper] [Patent]** Lightweight Countermeasures Against Original Linear Code Extraction Attacks on a RISC-V Core, HOST, 2023 (in collaboration with **Texplained**)
- Developed a Makefile-based environment for running Verilator simulations with assertions, and for launching Vivado FPGA design flows.
- Taught Computer Architecture and Digital Design to master's students.

• Engineering School Mines de St-Etienne 3 years

Award for academic excellence and involvement in associations.

Microelectronics: ASIC | FPGA | Manufacturing processes

- Hardware security: Laser fault injection, Electromagnetic Side-Channel Attacks

Computer science: Algorithmics | Machine Learning | TCP/IP Architecture

- Software security: Practical attacks (sniffing/man in the middle)

• Preparatory class : Mathematics & Physics 2 years

Intensive preparation for the national competitive entrance to leading French engineering schools

- Solved the traveling salesman problem using an "escargot shell" approach.

Professional experiences

• Assistant Hardware Security Engineer 6 months

Lightweight Control Flow integrity against FIA

- Researched the state-of-the-art solutions (Lock step/Watchdog).
- Developed a BB signature solution for the SODOR core (Berkley) on an FPGA.



• National RISC-V Contest 6 months

RISC-V core optimization

- Reduced program execution time without increasing silicon area.



• Research Internship in Biomechanics 5 months

Optimization of the violinist's movement

- Optimized the violinist's movement with the "Optimal control approach" in order to minimize muscular fatigue.
- Developed a generic "Optimal control" tool (Python/C++).



• Assistant Method Engineer 3 months

- Designed a defect detection with 3D Vision and probing equipment.



Volunteering experiences

• President of a Student Network Association "MINITEL" 1 year

Network Wi-Fi/Ethernet | Computer science | Events

- Managed an assorted team of 20 students
- Middleman between students and school (paid by the school)

Skills

Microelectronics

Vivado Xilinx	Modelsim Mentor Graphic	Verilator Veripool
Virtuoso Cadence	GTKWave Open Source	Synopsys Synopsys

Tools

Vim	Git	VS Code
Latex	Drawio	Inkscape

Programming

Python	C/Caml	C++
VHDL	S.Verilog	Makefile

Engineering & Research

Linux environment
Bibliographic research
Project management

Most visited website

Stack Overflow

Languages

French - **native** English - **C1**

Hobbies

Sport

Trek | Mountain bike | Kayak
(for long-term trips)

Environment

Climate Fresk | Facilitator

Culture

Stage acting

Science

C Genial contest (Popularization)